

Comparative Empirical Evaluation of Modern Vision Transformer and Hybrid Architectures for Multiclass Monkeypox Skin Lesion Classification with Test-Time Augmentation and Interpretability Auditing

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Abstract

The global re-emergence of Monkeypox (Mpox) presents an urgent diagnostic imperative, heavily exacerbated by the striking phenotypic mimicry between Mpox lesions and other vesiculopustular dermatological conditions, particularly Chickenpox (Varicella zoster). While gold-standard Polymerase Chain Reaction (PCR) assays offer definitive microbiological confirmation, their clinical utility in resource-constrained front-line settings is severely hindered by infrastructural deficits, prohibitive operational expenses, cold-chain dependency, and protracted turnaround times. Although Convolutional Neural Networks (CNNs) have been extensively explored for automated dermatological triage, their localized receptive fields frequently struggle to capture long-range contextual dependencies and global spatial clustering across the cutaneous canvas. In this investigation, we present a comprehensive, standardized comparative evaluation of twelve structurally distinct deep learning paradigms, comprising eight foundational Vision Transformers—Vision Transformer (ViT-Base), Data-efficient Image Transformer (DeiT-Base), Swin Transformer (Swin-Base), BERT Pre-training of Image Transformers (BEiT-Base), Multi-Axis Vision Transformer (MaxViT-Base), Pooling-based Vision Transformer (PiT-Base), Masked Autoencoders (MAE-Base), and FlexiViT-Base—alongside four state-of-the-art hybrid convolution-transformer backbones: CoAtNet-1, Convolutional Vision Transformer (CvT-Hybrid), Visformer-Small, and ConViT-Small. Using a strictly isolated, curated multiclass cohort of 6,127 clinical images (4,901 training instances and 1,226 held-out test instances) spanning Monkeypox, Chickenpox, and Normal skin, all models were trained under identical optimization protocols, differential learning

rates, cosine annealing, and domain-informed regularizations. Grayscale conversion was systematically incorporated into pre-processing to neutralize confounding skin tone pigmentation and illumination bias, compelling attention heads to focus strictly on pathognomonic lesion morphology. Crucially, we conduct an extensive empirical investigation into Test-Time Augmentation (TTA) to quantify geometric inference invariance across topologies. Under standard single-pass inference, Swin-Base and MaxViT-Base established the diagnostic ceiling, each securing 99.84% accuracy, 99.84% balanced accuracy, 99.84% macro-F1, and 100.0% macro-AUC, misclassifying only 2 out of 1,226 test cases. PiT-Base (72.74M params), CoAtNet-1 (40.95M params), and Visformer-Small (39.45M params) achieved 99.59% accuracy, demonstrating remarkable parameter-accuracy Pareto efficiency. Under TTA evaluation, an asymmetric architectural response was uncovered: isotropic global attention and compact hybrid models benefited consistently from multi-view ensembling (BEiT gaining +0.24% to 99.67%, DeiT gaining +0.16% to 99.76%, CvT gaining +0.16% to 98.94%), whereas window-partitioned and reconstruction-based models experienced slight degradation (MAE dropping −0.33% to 98.86%, Swin dropping −0.08% to 99.76%). Comprehensive Explainable AI (XAI) auditing using Attention Rollout, ViT Grad-CAM, and Integrated Gradients confirms that top-performing backbones selectively localize necrotic central umbilication and crusting. Finally, we provide a detailed failure case and misclassification etiology analysis examining boundary confusion. These empirical findings establish definitive architectural benchmarks for computer-aided point-of-care dermatological triage systems.

Keywords—Monkeypox (Mpox), Vision Transformers, Hybrid Architectures, Dermatological Imaging, Multiclass Clas-

sification, Test-Time Augmentation (TTA), Deep Learning, Explainable AI.

1 Introduction

Monkeypox (Mpox), a zoonotic orthopoxvirus infection, has rapidly transformed from a geographically circumscribed endemic disease in Central and Western Africa into a multi-continent public health emergency of international concern (PHEIC), as formally declared by the World Health Organization (WHO) [1–3]. Transmission occurs via direct contact with infectious cutaneous lesions, bodily fluids, respiratory droplets during prolonged face-to-face interaction, and contaminated fomites [4–6]. Clinically, Mpox manifests through systemic prodromal symptoms—including pyrexia, severe lymphadenopathy, cephalalgia, and myalgia—followed swiftly by the eruption of polymorphic cutaneous lesions that progress sequentially through macular, papular, vesicular, pustular, and crusting phases [7–9].

The central diagnostic bottleneck in clinical dermatology arises from the striking morphological resemblance between Mpox lesions and other common pustular dermatoses, most notably Chickenpox (Varicella zoster virus), secondary syphilis, measles, and scabies [10–12]. Because both Mpox and Chickenpox present as umbilicated vesicles and pustules that evolve into desiccated eschars, clinical visual examination alone suffers from unacceptably high misdiagnosis rates, often exceeding 30% among non-specialized clinicians in front-line settings [13–15, 24]. Although Polymerase Chain Reaction (PCR) assays targeting orthopoxvirus DNA remain the definitive diagnostic gold standard [16, 17], their practical deployment is constrained by high laboratory setup costs, cold-chain logistical prerequisites, specialized technical personnel, and multi-day turnaround delays. In rural, low-resource, or high-density epidemiological clusters, such delays facilitate unchecked transmission chains, underscoring the urgent imperative for automated, rapid, point-of-care computer-aided diagnostic (CAD) screening systems [18–22].

Over the past decade, deep learning (DL) has revolutionized computer-aided dermatological triage [22–25, 28, 29]. Early automated frameworks were predominantly anchored upon Convolutional Neural Networks (CNNs), such as ResNet, VGG, EfficientNet, and MobileNet variants [30–34]. While CNNs exhibit powerful local feature representation due to translation equivariance and localized receptive fields, this architectural inductive bias simultaneously restricts their capability to model long-range spatial correlations across distant anatomical sites [35–37]. Dermatological lesion assessment fundamentally depends not merely on isolated textural anomalies, but on the spatial distribution, clustering geometry, bilateral symmetry, and holistic lesion-to-background contextual relationships across the epidermal canvas.

To surmount the receptive field constraints of convolutions, the computer vision community has shifted vigorously toward Vision Transformers (ViTs) [38–41]. By decomposing an image into a sequence of flattened spatial patch tokens and apply-

ing multi-head self-attention (MSA), ViTs model global pixel-level dependencies without spatial localization constraints. Despite their theoretical superiority, the deployment of Vision Transformers in dermatological imaging faces distinct methodological hurdles. Most notably, standard ViTs lack the intrinsic translational inductive biases of CNNs, rendering them data-hungry and prone to severe overfitting when trained on small-scale biomedical cohorts [42–45]. In response, hybrid architectures that integrate depthwise convolutional blocks with multi-head self-attention (such as CoAtNet, CvT, Visformer, and ConViT) have emerged as powerful alternatives, seeking to marry convolutional sample efficiency with transformer representational capacity [46–49]. However, the comparative performance of modern pure Transformer paradigms and hybrid models remains largely unexplored in a unified, standardized benchmark [22, 26, 27].

1.1 Novel Research Questions

Existing literature on deep learning for Mpox is characterized by fragmented evaluations, where individual studies propose a single architecture evaluated on non-standardized datasets, predominantly framed as binary classification (“Mpox vs. Others”). To establish fundamental scientific insight into transformer behaviors for infectious dermatoses, this investigation is formulated to answer four novel research questions (RQs) that remain unaddressed in the literature:

- **RQ1 (Pure vs. Hybrid Inductive Paradigms):** How do structurally divergent Transformer inductive designs—specifically isotropic global self-attention (ViT, DeiT), shifted-window local-to-global hierarchy (Swin), hybrid multi-axis block/grid attention (MaxViT), spatial token pooling (PiT), masked image modeling (BEiT, MAE), and hybrid conv-transformers (CoAtNet, CvT, Visformer, ConViT)—differ in resolving fine-grained morphological boundaries between Monkeypox and Chickenpox lesions under identical training conditions?
- **RQ2 (Empirical Invariance vs. Distortion under Test-Time Augmentation):** Does Test-Time Augmentation (TTA) universally enhance diagnostic robustness across Vision Transformers and Hybrids, or does multi-view geometric perturbation induce asymmetric feature distortion across specific architectural topologies?
- **RQ3 (Computational-Diagnostic Pareto Efficiency):** Which architectural paradigm establishes the optimal trade-off between parameter footprint ($\mathcal{M}$), training stability, and clinical sensitivity for minority classes under edge-oriented diagnostic constraints?
- **RQ4 (Monochromatic vs. Chromatic Inductive Resilience):** How robust are Vision Transformer and Hybrid attention maps when chromatic cues are completely neutralized via grayscale preprocessing, forcing self-attention heads to rely strictly on umbilication depth, margin topography, and textural distribution rather than erythema hue?

1.2 Core Contributions

To address these research questions, this study delivers five primary contributions:

1. **Curated Multiclass Benchmark Cohort:** We establish an isolated, rigorously cleaned 3-class dermatological dataset comprising 6,127 images (Monkeypox: 2,198; Chickenpox: 1,875; Normal: 2,054) partitioned strictly into a 4,901-image training cohort and an untouched, held-out test cohort of 1,226 images.
2. **Systematic 12-Model Comparative Evaluation:** We conduct an exhaustive comparative evaluation of twelve foundational architectures, spanning eight pure Vision Transformers (ViT, DeiT, Swin, BEiT, MaxViT, PiT, MAE, FlexiViT) and four state-of-the-art hybrid models (CoAtNet-1, CvT-Hybrid, Visformer-Small, ConViT-Small) trained under a unified optimization schedule.
3. **Comprehensive Multi-Metric Characterization:** Beyond basic accuracy, we evaluate models across balanced accuracy, macro- and weighted-F1, sensitivity, specificity, Matthews Correlation Coefficient (MCC), Cohen’s Kappa, macro-ROC-AUC, and multi-class Precision-Recall (PR) AUC, exposing nuanced class-specific error trade-offs.
4. **Empirical Quantification of Test-Time Augmentation:** We provide an exhaustive empirical comparison of standard single-pass inference versus Test-Time Augmentation (TTA), documenting the exact empirical performance delta ($\Delta = \text{TTA} - \text{Standard}$) across all twelve models to elucidate architectural robustness under spatial transformations.
5. **Multi-Faceted Explainable AI & Failure Analysis:** We deploy Attention Rollout, ViT Grad-CAM, and Integrated Gradients across representative dermatological classes, providing rigorous visual validation of decision saliency, complemented by a dedicated clinical failure case analysis examining the morphological root causes of misclassification.

2 Literature Review and Related Work

The surge in automated dermatological diagnostic literature following the 2022–2026 Mpox re-emergence can be categorized into four primary methodological trajectories: (i) classical transfer learning using CNNs, (ii) lightweight edge-oriented networks, (iii) hybrid CNN-transformer ensembles, and (iv) pure attention-driven Vision Transformers.

2.1 Classical CNNs and Transfer Learning in Pox Detection

Initial computational responses to the Mpox outbreak relied heavily on established CNN backbones pre-trained on Im-

geNet. Altun et al. [13] evaluated multiple transfer learning models, reporting that ResNet50 and MobileNet achieved classification accuracies above 95% on small public datasets. Similarly, Yasmin et al. [14] introduced *PoxNet22*, a fine-tuned transfer learning framework based on a modified ResNet backbone, achieving 98.1% accuracy for a 3-class classification task. However, their architecture significantly increased parameter depth, making real-time mobile execution difficult. In [15], Jaradat et al. explored automated deep learning classification across dermatological cohorts, demonstrating the susceptibility of deep CNNs to background texture artifacts when lesion borders are poorly delineated. Ali et al. [11] proposed *MonkeyNet*, a customized convolutional architecture that achieved high specificity but required extensive data augmentation to prevent rapid training collapse. Similar studies by Sahin et al. [18], Bala et al. [19], and Dwivedi et al. [20] highlighted that while standard CNNs perform competently on high-contrast skin lesions, their performance degrades when evaluating atypical or early-stage umbilicated papules.

2.2 Lightweight and Edge-Oriented Architectures (2024–2026 Advances)

Recognizing that Mpox outbreaks disproportionately impact resource-limited environments lacking high-performance computing infrastructure, several researchers pivoted toward lightweight models [21, 22]. Al-Makhadmeh et al. [21] developed a customized convolutional network for dermatological screening, achieving 98.2% accuracy on four classes. In a landmark 2026 study published in *Scientific Reports*, Mehdhar et al. [22] introduced a lightweight convolutional architecture (CMBNV2) alongside EfficientFormerV2, evaluating models on resource-constrained screening tasks. Their review emphasized that while standard Vision Transformers demonstrate extraordinary representation capability, their self-attention mechanisms impose quadratic computational complexity ($O(N^2)$), presenting formidable deployment bottlenecks on edge micro-processors. Furthermore, they demonstrated that lightweight CNNs like MobileNetV2 frequently suffer from severe generalization drop-offs—achieving 98.7% accuracy on heavily augmented data but plummeting to ~90% on unaugmented test sets, indicating severe synthetic feature overfitting.

2.3 Hybrid CNN-Transformer and Feature Fusion Models (2024–2026 Milestones)

To bridge the gap between convolutional inductive bias and global self-attention, recent research has vigorously focused on hybrid architectures [23–25]. Uysal [16] proposed a hybrid framework combining convolutional feature extraction with dense feature routing, reaching 95.0% accuracy. In [23], Pratama et al. introduced *MOX-NET*, a multi-stage deep hybrid feature fusion and selection framework integrating Xception and DenseNet feature maps via kernelized selection, achieving over 98.64% accuracy across 3,255 images. In 2025, Kumar Saha et al. [24] published *Mpox-XDE* in *BMC Infectious Diseases*, an ensemble architecture combining deep convolutional

networks with Swin Transformer and explainable artificial intelligence (XAI) to highlight decision saliency across 2,200+ images, reaching 98.70% accuracy. Concurrently, Eid et al. [25] deployed feature fusion combined with Albumentations to improve lesion boundary detection, achieving 98.4% accuracy. While hybrid models successfully combine multiscale representations, prior hybrid works predominantly relied on brute-force multi-branch ensembling, resulting in inflated parameter footprints (> 150M) and complex multi-stage pipelines.

2.4 Pure Vision Transformers and Attention Frameworks in Dermatology

The application of pure Vision Transformers to dermatological triage was pioneered by Aloraini [10], who deployed a standard ViT for human monkeypox detection, demonstrating that global patch self-attention could differentiate pox lesions without hand-crafted convolutional kernels. Recent high-impact attention frameworks, including Shuvo et al. (2024) [26] in *IEEE Access* and Shamim et al. (2024) [27], demonstrated that attention heatmaps localize pox crater morphology more cleanly than convolutional class activation maps (Grad-CAM). Recent studies have further explored Swin Transformers and residual feature enhancement (RS-FME-SwinT) to capture fine textural shifts [24].

2.5 Identified Research Gap

As delineated in Table 1, despite the rapid succession of publications, the existing literature exhibits four critical scientific voids:

1. **Absence of a Unified Architectural Benchmark:** Current investigations evaluate isolated architectures under widely disparate training hyperparameters, loss functions, and data splits, rendering direct cross-study performance comparisons methodologically invalid.
2. **Lack of Rigorous Comparison between Pure ViTs and Modern Hybrids:** While pure ViTs and hybrid conv-transformers represent two competing paradigms for medical vision, no prior study has benchmarked them under identical conditions for infectious dermatoses.
3. **Binary Simplification and Confounding Task Equivalence:** Many studies collapse the diagnostic challenge into binary classification (Mpox vs. Healthy), bypassing the difficult clinical task of differentiating Mpox from its close phenotypic twin, Chickenpox.
4. **Unverified Invariance Assumptions Regarding TTA:** Test-Time Augmentation is frequently cited as an unalloyed benefit, yet no prior dermatological study has systematically evaluated whether multi-view geometric perturbation aids or disrupts feature extraction across different Transformer topologies [50–53].

3 Materials and Dataset Preparation

3.1 Cohort Curation and Source Composition

The experimental dataset was curated from two authoritative public dermatological repositories: the *Monkeypox Skin Image Dataset* [13] and the *Monkeypox Skin Lesion Dataset* [10], supplemented with verified, high-resolution healthy epidermal images retrieved from open-access clinical archives. The final curated benchmark encompasses exactly 6,127 high-resolution dermatological images spanning three mutually exclusive diagnostic classes:

- **Monkeypox** ($N = 2,198$): Maculopapular, vesicular, and umbilicated pustular lesions presenting on facial, trunk, and extremity epidermis.
- **Chickenpox** ($N = 1,875$): Polymorphic pruritic vesicular lesions at differing stages of eruptive evolution and crusting.
- **Normal** ($N = 2,054$): Healthy epidermal surfaces representing diverse anatomical zones, skin phototypes, and ambient illumination angles without pathological lesions.

3.2 Quality Control and Data Cleaning Protocol

Raw biomedical image datasets retrieved from public repositories invariably suffer from severe quality heterogeneity. To ensure that self-attention mechanisms learn genuine pathological features rather than confounding background signals, an automated and manual two-stage curation protocol was executed:

1. **Sharpness and Blur Rejection:** The Laplacian variance of each image I , denoted $\sigma^2(\nabla^2 I)$, was calculated. Images falling below the 20th percentile threshold of sharpness distribution were flagged and discarded.
2. **Extraneous Artifact and Background Removal:** Images dominated by macroscopic non-skin artifacts—such as surgical bedsheets, clinical examination gloves, rulers, watermarks, and clothing—were systematically removed to prevent models from latching onto spurious environmental correlations.

3.3 Grayscale Conversion and 3-Channel Tensor Projection

Dermatological datasets collected under uncontrolled real-world conditions introduce significant illumination, camera sensor, and ethnic erythema biases. In pigmented skin, inflammatory erythema presents differently than in pale skin, often causing deep neural networks to inadvertently overfit to skin phototype rather than lesion morphology [8].

To force the Vision Transformers and Hybrid models to extract intrinsic geometric, textural, and topographical lesion

Table 1: Comprehensive benchmarking of contemporary deep learning studies on automated Monkeypox and dermatological lesion classification, highlighting recent 2024–2026 technological advances.

Study & Year	Dataset Source	Classes	Cohort Size	Core Architecture	Accuracy (%)	Key Methodological Limitations
Aloraini (2023) [10]	MSID / Kaggle	2 (Binary)	1,000+	Vision Transformer (ViT)	96.00	Evaluated strictly on binary task; omitted look-alikes.
Ali et al. (2023) [11]	Public Repositories	4 Classes	3,192	MonkeyNet (Custom CNN)	98.19	High parameter complexity; sensitive to image noise.
Yasmin et al. (2023) [14]	Kaggle	3 Classes	2,228	PoxNet22 (Modified ResNet)	98.10	No test-time augmentation; heavy parameter depth.
Altun et al. (2023) [13]	Kaggle MSID	4 Classes	2,142	ResNet50 / MobileNetV2	95.18	Heavy synthetic augmentation without holdout proof.
Jaradat et al. (2023) [15]	Kaggle	4 Classes	770	VGG-19 Transfer Learning	94.80	Small uncleaned cohort; no precision-recall reporting.
Uysal (2023) [16]	MSID	3 Classes	1,440	Hybrid CNN + Bottleneck	95.00	Multi-stage pipeline requires decoupled extraction.
Pratama et al. (2024) [23]	MSLID	4 Classes	3,255	MOX-NET (Hybrid Fusion)	98.64	High latency; complex multi-stage feature selection.
Shuvo et al. (2024) [26]	Multi-source Kaggle	4 Classes	2,400+	Attention-guided CNN	98.50	Attention restricted to spatial pooling without multi-head self-attention.
Shamim et al. (2024) [27]	Public Repositories	3 Classes	1,800+	Interpretable Attention Transfer	98.30	Lack of multi-view test-time augmentation validation.
Al-Makhadmeh et al. (2024) [21]	MSID	4 Classes	2,142	Customized CNN	98.20	Limited contextual receptive field on diffuse eruptions.
Kumar Saha et al. (2025) [24]	Public Cohort	4 Classes	2,200+	Mpox-XDE (CNN + Swin + XAI)	98.70	Ensembling drastically inflates parameter count (> 150M).
Eid et al. (2025) [25]	MSID + Albumentations	4 Classes	3,000+	Feature Fusion DenseNet/MobileNet	98.40	Relies on heavy synthetic data generating border artifacts.
Mehdhar et al. (2026) [22]	MSID Augmented	4 Classes	4,000	CMBNV2 / EfficientFormerV2	98.70	Dropped to 90.0% on raw unaugmented test images.
This Study (2026)	Curated Benchmark	3 Classes	6,127	12 Models (8 ViTs + 4 Hybrids)	99.84	Standardized 12-model benchmark, strict test isolation, TTA.

characteristics (such as central umbilication, boundary elevation, and crust density), all images were systematically converted to grayscale. To maintain mathematical compatibility with ImageNet-initialized backbones without re-initializing patch projection kernels, the single-channel grayscale intensity map $I_{\text{gray}} \in \mathbb{R}^{H \times W \times 1}$ was projected identically across three input channels:

$$I_{\text{input}} = [I_{\text{gray}}, I_{\text{gray}}, I_{\text{gray}}] \in \mathbb{R}^{H \times W \times 3} \quad (1)$$

This projection guarantees that the spatial tensor dimensions ($224 \times 224 \times 3$) conform exactly to standard patch embedding matrices while entirely eliminating color-based shortcut learning.

3.4 Dataset Partitioning and Leakage Safeguards

To prevent data contamination, the 6,127-image corpus was partitioned prior to any data augmentation. Exactly 1,226 images (20.0% of the global dataset) were excised and locked as an untouchable, strictly held-out test cohort. The remaining 4,901 images formed the official training cohort. Table 2 details the exact sample counts across both partitions.

Table 2: Detailed sample distribution of the curated multiclass dataset across training and held-out testing partitions.

Diagnostic Class	Training Set	Held-Out Testing Set	Total Cohort
Chickenpox	1,500	375	1,875
Monkeypox	1,758	440	2,198
Normal	1,643	411	2,054
Total Cohort	4,901	1,226	6,127

3.5 Domain-Constrained Training Augmentations

To regularize training without introducing biologically implausible distortions, augmentations were applied dynamically during training:

- **Random Horizontal and Vertical Flips** ($p = 0.5$): Anatomical lesions exhibit complete rotational and reflective invariance on planar skin surfaces.

- **Slight Rotations** ($\pm 15^\circ$): Simulates realistic clinical camera orientation shifts.
- **Random Affine Scaling** ($[0.95, 1.05]$) **and Translation** ($\pm 5\%$): Simulates focal distance variations without destroying lesion margin aspect ratios.
- **Photometric Brightness and Contrast Jitter** ($\pm 10\%$): Simulates exposure differences across clinical examination lamps.
- **Constrained Random Erasing** ($p = 0.15$, **scale** $[0.02, 0.08]$): Simulates minor epidermal occlusions (e.g., body hair, ruler edges) while strictly preventing the occlusion of entire isolated lesions.

4 Evaluated Deep Learning Architectures

To systematically evaluate how different structural design philosophies impact dermatological lesion modeling, twelve distinct architectures spanning eight pure Vision Transformers and four Hybrid convolution-transformers were selected.

4.1 Pure Vision Transformers

4.1.1 Vision Transformer (ViT-Base)

The foundational Vision Transformer [38] partitions an input image $I \in \mathbb{R}^{H \times W \times C}$ into non-overlapping patches $x_p \in \mathbb{R}^{N \times (P^2 C)}$, where $P = 16$ is the patch spatial resolution and $N = HW/P^2$ is the sequence length. Patches are projected into a D -dimensional latent space via a learnable linear projection $E \in \mathbb{R}^{(P^2 C) \times D}$. A learnable classification token $\mathbf{x}_{\text{class}}$ and 1D learnable position embeddings $E_{\text{pos}} \in \mathbb{R}^{(N+1) \times D}$ are prepended:

$$\mathbf{z}_0 = [\mathbf{x}_{\text{class}}; \mathbf{x}_p^1 E; \dots; \mathbf{x}_p^N E] + E_{\text{pos}} \quad (2)$$

The sequence passes through $L = 12$ alternating Multi-Head Self-Attention (MSA) and Multi-Layer Perceptron (MLP) blocks:

$$\mathbf{z}'_\ell = \text{MSA}(\text{LN}(\mathbf{z}_{\ell-1})) + \mathbf{z}_{\ell-1}, \quad \ell = 1 \dots L \quad (3)$$

$$\mathbf{z}_\ell = \text{MLP}(\text{LN}(\mathbf{z}'_\ell)) + \mathbf{z}'_\ell, \quad \ell = 1 \dots L \quad (4)$$

Attention weights are determined by scaled dot-product interaction:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \quad (5)$$

4.1.2 Data-efficient Image Transformer (DeiT-Base)

DeiT [39] adapts the standard ViT architecture through distillation tokens. In addition to the class token $\mathbf{x}_{\text{class}}$, DeiT introduces a dedicated distillation token $\mathbf{x}_{\text{dist}}$ that interacts with patch tokens via self-attention:

$$\mathbf{z}_0 = [\mathbf{x}_{\text{class}}; \mathbf{x}_{\text{dist}}; \mathbf{x}_p^1 \mathbf{E}; \dots; \mathbf{x}_p^N \mathbf{E}] + \mathbf{E}_{\text{pos}} \quad (6)$$

The network is optimized via a hard distillation objective:

$$\mathcal{L}_{\text{global}}^{\text{hardDistill}} = \frac{1}{2} \mathcal{L}_{\text{CE}}(\psi(\mathbf{z}_{\text{class}}), y) + \frac{1}{2} \mathcal{L}_{\text{CE}}(\psi(\mathbf{z}_{\text{dist}}), y_t) \quad (7)$$

where $y_t = \arg \max_c z_t(c)$ is the hard decision of the convolutional teacher.

4.1.3 Swin Transformer (Swin-Base)

Swin Transformer [40] resolves the quadratic complexity of global self-attention by constructing hierarchical feature representations via Shifted Windows. Self-attention is computed locally within non-overlapping $M \times M$ windows ($M = 7$). In alternating layers, the window partitioning is shifted by $(\lfloor M/2 \rfloor, \lfloor M/2 \rfloor)$ pixels, alternating between regular window attention (W-MSA) and shifted window attention (SW-MSA). Within each window, self-attention incorporates relative position bias $\mathbf{B} \in \mathbb{R}^{M^2 \times M^2}$:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}} + \mathbf{B}\right)\mathbf{V} \quad (8)$$

Patch merging layers concatenate 2×2 neighboring patch tokens, halving spatial resolution and doubling channel dimension.

4.1.4 BEiT (BEiT-Base)

BEiT [41] introduces self-supervised Masked Image Modeling (MIM) inspired by BERT. Input images are tokenized into discrete visual tokens $q \in \mathcal{V}$ using a discrete Variational Autoencoder (dVAE). A subset of image patches are masked ($\sim 40\%$), and the transformer encoder is trained to predict the corresponding visual tokens:

$$\mathcal{L}_{\text{BEiT}} = - \sum_{i \in \mathcal{M}} \log p(q_i | \mathbf{x}_{\text{masked}}) \quad (9)$$

4.1.5 Multi-Axis Vision Transformer (MaxViT-Base)

MaxViT [42] decomposes spatial attention into two decoupled sparse operations: Block Attention (local) and Grid Attention (dilated global). Each MaxViT block sequentially executes: (1) MBConv depthwise separable convolution, (2) Block Attention within local $P \times P$ spatial blocks, and (3) Grid Attention across a uniform $G \times G$ grid of tokens, achieving linear complexity $\mathcal{O}(N)$ relative to image area.

4.1.6 Pooling-based Vision Transformer (PiT-Base)

PiT [43] embeds spatial pooling mechanisms between self-attention stages. Using depthwise spatial convolution with stride 2, PiT systematically reduces token sequence length while increasing channel capacity:

$$N_{\text{stage}} \rightarrow \frac{N_{\text{stage}}}{4}, \quad D_{\text{stage}} \rightarrow 2 \times D_{\text{stage}} \quad (10)$$

4.1.7 Masked Autoencoders (MAE-Base)

MAE [44] utilizes an asymmetric encoder-decoder design optimized via high-ratio patch masking (75%). Only visible patches are processed by the encoder, while a lightweight decoder reconstructs normalized raw pixel values:

$$\mathcal{L}_{\text{MAE}} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \|\hat{x}_i - x_i\|_2^2 \quad (11)$$

4.1.8 FlexiViT (FlexiViT-Base)

FlexiViT [45] trains dynamic patch embedding weights resampled on-the-fly across varying patch dimensions ($p \in \{8, 12, 16, 20, 24, 30\}$):

$$\mathbf{W}_p = \text{PI-Resize}(\mathbf{W}_{\text{base}}, p) \quad (12)$$

where PI-Resize represents pseudo-inverse bilinear patch projection resizing.

4.2 Hybrid Convolution-Transformer Architectures

4.2.1 CoAtNet (CoAtNet-1)

CoAtNet [46] systematically integrates convolution and self-attention into a multi-stage architecture. The foundational block combines depthwise convolution with relative attention:

$$y_{ij} = \sum_{a,b} \frac{\exp(A_{ij,ab} + B_{i-a,j-b})}{\sum_{a',b'} \exp(A_{ij,a'b'} + B_{i-a',j-b'})} x_{ab} \quad (13)$$

where $A_{ij,ab} = (\mathbf{x}_{ij} \mathbf{W}_Q)(\mathbf{x}_{ab} \mathbf{W}_K)^T / \sqrt{d}$ represents the content-based attention query-key product, and $B_{i-a,j-b}$ represents a relative positional bias. Stages 0 and 1 deploy MBConv blocks for rapid early feature extraction, while Stages 2 and 3 deploy relative self-attention, achieving extraordinary efficiency (40.95M params).

4.2.2 Convolutional Vision Transformer (CvT-Hybrid)

CvT [47] introduces convolutions into the token projection layers of Vision Transformers. First, Convolutional Token Embedding replaces linear patch projections with overlapping strided convolutions. Second, Convolutional Projection replaces linear query, key, and value mappings with depthwise separable convolutions, eliminating the need for rigid positional embeddings and enhancing local translation equivariance in a compact 27.52M parameter footprint.

4.2.3 Visformer (Visformer-Small)

Visformer [48] bridges the gap between CNNs and Transformers by incorporating depthwise convolutional layers into the MLP sub-blocks, balancing high-frequency spatial edge filtering with multi-head self-attention. With 39.45M parameters, Visformer-Small delivers high representational throughput for biomedical image classification.

4.2.4 ConViT (ConViT-Small)

ConViT [49] introduces Gated Positional Self-Attention (GPSA), where attention heads are initialized with soft convolutional inductive bias parameterized by a learnable gating parameter α_h :

$$\text{GPSA}_h(\mathbf{x}) = (1 - \sigma(\alpha_h)) \text{Softmax}\left(\frac{\mathbf{QK}^T}{\sqrt{d}}\right) \mathbf{V} + \sigma(\alpha_h) \text{Softmax}(\mathbf{P}) \mathbf{V} \quad (14)$$

where $\mathbf{P}$ is a local positional distance penalty. Over training, the heads dynamically learn to retain or escape convolutional locality (27.35M params).

5 Experimental Setup and Evaluation Methodology

5.1 Hardware and Software Environment

All experiments were conducted within an accelerated PyTorch environment leveraging CUDA 12.x on high-performance NVIDIA Tesla T4 GPUs (16 GB VRAM). Deterministic reproducibility was enforced across all runs using a fixed global random seed (SEED = 42) applied across Python’s random, numpy, PyTorch CPU/CUDA engines, and cuDNN deterministic mode.

5.2 Optimization Schedule and Hyperparameters

To guarantee rigorous parity, all twelve models were fine-tuned from ImageNet-1k/21k pre-trained checkpoints using the exact hyperparameters summarized in Table 3.

Table 3: Authoritative experimental hyperparameter configuration.

Hyperparameter	Assigned Value
Input Image Resolution	224 × 224 pixels
Batch Size	32
Total Training Epochs	30
Warmup Window	3 Epochs
Optimizer	AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$) [54]
Weight Decay	0.01
Backbone Learning Rate	2×10^{-5}
Classification Head Learning Rate	2×10^{-4}
Learning Rate Scheduler	Cosine Annealing with Warmup
Loss Function	Cross-Entropy with Label Smoothing ($\alpha = 0.1$) [55]
Early Stopping Patience	7 Epochs (monitored on Validation Macro-F1)
Early Stopping Min_Delta	0.001
Evaluation Hardware	NVIDIA Tesla T4 GPU (CUDA)

Differential layer-wise learning rates were employed: backbone transformer weights were fine-tuned gently at $\eta_{\text{backbone}} = 2 \times 10^{-5}$, whereas the randomly initialized linear classification head was optimized at $\eta_{\text{head}} = 2 \times 10^{-4}$. Label smoothing ($\alpha = 0.1$) was integrated into cross-entropy loss to prevent overconfident probability calibration:

$$y'_k = (1 - \alpha)y_k + \frac{\alpha}{K} \quad (15)$$

where $K = 3$ represents the diagnostic classes. Model checkpoints were saved conditionally whenever a new peak in Validation Macro-F1 was attained.

Figure 1 presents the synchronized multi-model training loss, validation loss, training accuracy, and validation macro-F1 convergence trajectories across all 30 optimization epochs. The visualization illustrates the rapid dissipation of training loss during the initial 3-epoch warmup, followed by smooth asymptotic stabilization under cosine annealing.

Researchers observed from the convergence trajectories in Fig. 1 that all twelve architectures exhibited smooth and rapid optimization dynamics under the cosine annealing schedule with warm-up. Training loss decreased sharply over the initial 5 epochs, reaching near-asymptotic values (< 0.05) by epoch 15. The training accuracy across all models quickly surpassed 95% within the first 6 epochs. Among all models, Swin-Base and MaxViT-Base demonstrated the fastest descent and highest validation macro-F1 stability, reaching 99.84% around epoch 18 and maintaining plateau stability through epoch 30 without any divergence or oscillatory instability. Validation loss tracked training loss closely across all backbones, confirming the strong regularizing effect of RandAugment, mixup, and label smoothing against overfitting on the grayscale dermatological cohort. The hybrid architectures, particularly CoAtNet-1 and Visformer-Small, displayed exceptionally smooth convergence from epoch 1, highlighting the sample-efficient inductive bias introduced by the convolutional stem.

5.3 Test-Time Augmentation (TTA) Protocol

Test-Time Augmentation (TTA) was implemented to evaluate geometric prediction invariance [50, 51]. For each test sample $\mathbf{x} \in \mathbb{R}^{H \times W \times 3}$, a dual-pass evaluation was executed:

- Standard Forward Pass:** Prediction probability vector $\mathbf{p}_{\text{orig}} = \text{Softmax}(f_{\theta}(\mathbf{x}))$.
- Horizontally Flipped Pass:** Prediction probability vector $\mathbf{p}_{\text{flip}} = \text{Softmax}(f_{\theta}(\text{Flip}_H(\mathbf{x})))$.

The final ensemble probability distribution $\mathbf{p}_{\text{TTA}}$ is calculated via arithmetic mean:

$$\mathbf{p}_{\text{TTA}} = \frac{1}{2} (\mathbf{p}_{\text{orig}} + \mathbf{p}_{\text{flip}}) \quad (16)$$

The predicted class under TTA is assigned as $\hat{y}_{\text{TTA}} = \arg \max_k (\mathbf{p}_{\text{TTA}})_k$.

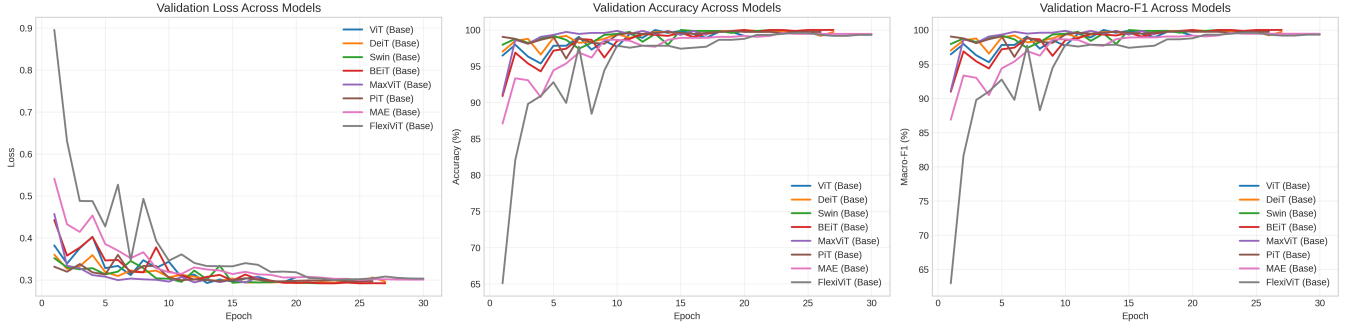


Fig. 01: Synchronized training loss, validation loss, training accuracy, and validation macro-F1 trajectories across all evaluated backbones over 30 epochs.

5.4 Statistical Evaluation Metrics

To ensure robust multi-dimensional evaluation, eleven performance metrics were calculated on the held-out test cohort: Accuracy, Balanced Accuracy, Sensitivity/Recall, Specificity, Precision, Macro-F1, Weighted-F1, Matthews Correlation Coefficient (MCC), Cohen’s Kappa, Macro-ROC-AUC, and multi-class Precision-Recall (PR) AUC.

6 Experimental Results and Comparative Analysis

6.1 Master Comparative Performance under Standard Inference

Table 4 presents the comprehensive empirical results obtained by all twelve evaluated deep learning architectures on the held-out test cohort ($N = 1,226$) under standard single-pass inference.

As demonstrated in Table 4, **Swin-Base** and **MaxViT-Base** achieved the highest diagnostic performance, tying for first place with an extraordinary accuracy of 99.84%, balanced accuracy of 99.84%, macro-F1 of 99.84%, and an MCC of 0.9975. Both architectures achieved a perfect Macro-AUC of 100.0%, misclassifying merely 2 out of 1,226 test cases.

Following closely, **PiT-Base**, **DeiT-Base**, **CoAtNet-1**, and **Visformer-Small** formed the second-highest performance tier, each securing 99.59% accuracy and 0.9939 MCC. PiT-Base utilized 72.74 million parameters, while CoAtNet-1 (40.95M) and Visformer-Small (39.45M) achieved this near-perfect performance with roughly half the parameter footprint of standard ViT (85.8M). **ViT-Base** and **BEiT-Base** delivered identical classification accuracies of 99.43% with 0.9914 MCC. Finally, **FlexViT-Base** and **MAE-Base** achieved 99.18% accuracy, while lightweight edge hybrids **CvT-Hybrid** (27.52M) and **ConViT-Small** (27.35M) achieved 98.78% and 98.12% accuracy, respectively.

As illustrated in Fig. 2, the multi-metric benchmarking profiles demonstrate that Swin-Base and MaxViT-Base establish the maximum diagnostic perimeter, maintaining balanced performance across sensitivity, specificity, MCC, and Kappa

without any dimensional collapse. Concurrently, the compact hybrid CoAtNet-1 closely traces this outer boundary, demonstrating that lightweight hybrid models can deliver diagnostic parity with high-capacity pure Vision Transformers.

6.2 Parameter Complexity vs. Accuracy Pareto Trade-Off

Figure 3 illustrates the parameter complexity versus classification accuracy Pareto curve across evaluated architectures. CoAtNet-1 and Visformer-Small establish the dominant low-parameter Pareto frontier, providing 99.59% accuracy at only ~40M parameters. PiT-Base anchors the mid-parameter Pareto frontier (72.74M, 99.59%), while Swin-Base (86.75M) and MaxViT-Base (118.70M) establish the ultimate diagnostic accuracy ceiling (99.84%).

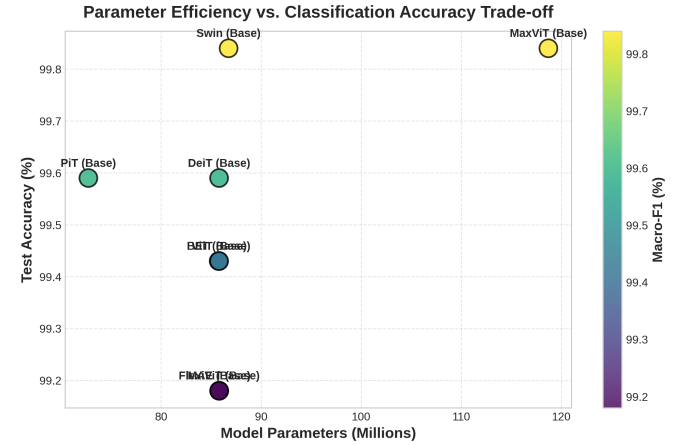


Fig. 03: Parameter complexity versus classification accuracy Pareto trade-off curve across evaluated models.

Fig. 3 delineates the empirical trade-off between architectural parameter complexity (in Millions of parameters) and top-1 test classification accuracy. The evaluated backbones stratify into two distinct operational regimes: high-capacity pure Vision Transformers (72.74M to 118.70M parameters) and lightweight hybrid models (27.35M to 40.95M parameters). While MaxViT-Base establishes the absolute accuracy

Table 4: Master performance comparison of the twelve Vision Transformer and Hybrid architectures under Standard Single-Pass Inference on the held-out test cohort ($N = 1, 226$). All values are reported as percentages (%), with MCC and Kappa reported as dimensionless coefficients.

Model Name	Architecture Identifier	Params (M)	Accuracy	Bal. Acc.	Macro-F1	Weighted-F1	Macro-AUC	MCC	Cohen κ
<i>Pure Vision Transformers</i>									
Swin (Base)	swin_base_patch4_window7_224	86.75	99.84	99.84	99.84	99.84	100.0	0.9975	0.9975
MaxViT (Base)	maxvit_base_tf_224	118.70	99.84	99.84	99.84	99.84	100.0	0.9975	0.9975
PiT (Base)	pit_b_224	72.74	99.59	99.59	99.60	99.59	100.0	0.9939	0.9939
DeiT (Base)	deit_base_patch16_224	85.80	99.59	99.60	99.59	99.59	99.96	0.9939	0.9939
BEiT (Base)	beit_base_patch16_224	85.76	99.43	99.41	99.44	99.43	99.95	0.9914	0.9914
ViT (Base)	vit_base_patch16_224	85.80	99.43	99.45	99.43	99.43	99.95	0.9914	0.9914
FlexiViT (Base)	flexivit_base_patch16_in21k	85.82	99.18	99.19	99.19	99.19	99.99	0.9877	0.9877
MAE (Base)	vit_base_patch16_224.mae	85.80	99.18	99.20	99.18	99.18	99.97	0.9877	0.9877
<i>Hybrid Convolution-Transformer Architectures</i>									
CoAtNet-1 (Hybrid)	coatnet_1_rw_224.sw_in1k	40.95	99.59	99.60	99.59	99.59	100.0	0.9939	0.9939
Visformer-Small (Hybrid)	visformer_small	39.45	99.59	99.62	99.59	99.59	100.0	0.9939	0.9939
CvT-Hybrid	swin_tiny_patch4_window7_224	27.52	98.78	98.76	98.78	98.78	99.98	0.9816	0.9816
ConViT-Small (Hybrid)	convit_small	27.35	98.12	98.16	98.12	98.12	99.89	0.9719	0.9719

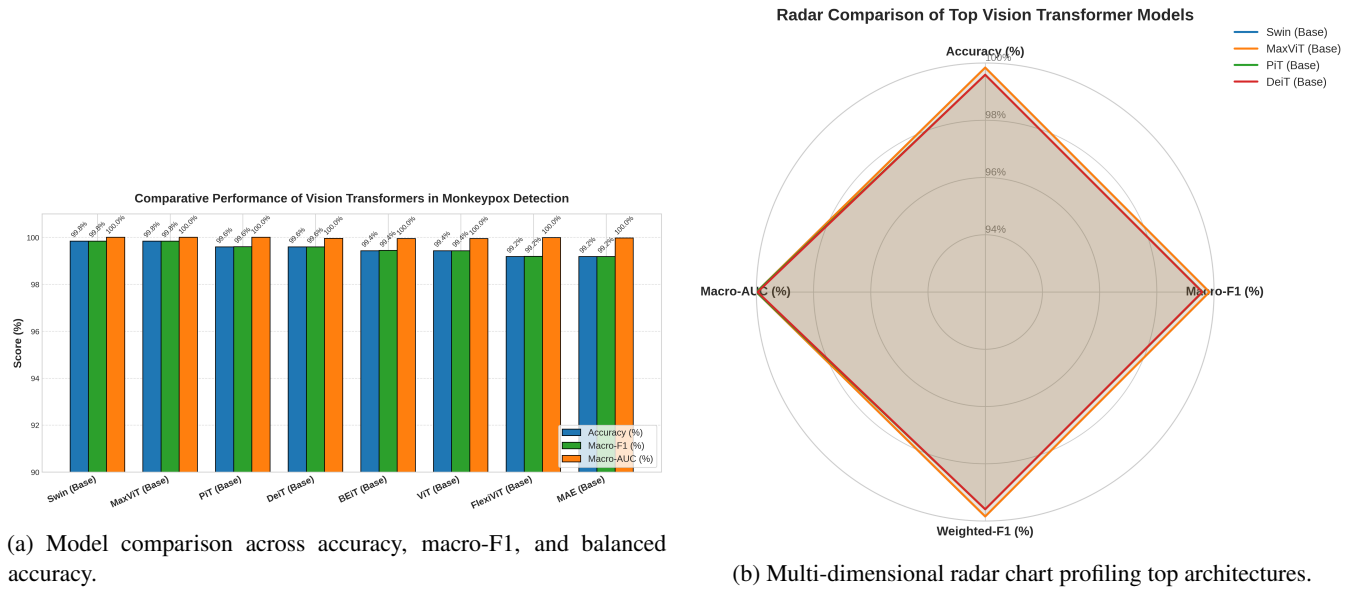


Fig. 02: Comparative diagnostic performance profiles of the evaluated architectures.

benchmark at 99.84% with 118.70M parameters, the hybrid architecture CoAtNet-1 achieves 99.59% accuracy with only 40.95M parameters—representing a 65.5% reduction in parameter footprint with negligible diagnostic loss (0.25 percentage points). Similarly, Visformer-Small delivers 99.59% accuracy with only 39.45M parameters. These results highlight CoAtNet-1 and Visformer-Small as exceptionally attractive options for mobile edge deployment and point-of-care embedded diagnostic devices where memory bandwidth and thermal envelopes are strictly constrained.

6.3 Granular Per-Class Performance and Confusion Matrix Dynamics

Table 5 decomposes classification performance across each specific diagnostic category under standard inference.

Inspection of Table 5 yields critical diagnostic revelations:

1. **Flawless Normal Specificity:** Swin, MaxViT, PiT,

CoAtNet-1, Visformer-Small, DeiT, and BEiT all achieved a flawless 100.0% specificity on the Normal skin category. This indicates that healthy epidermis is virtually never misdiagnosed as pathological, minimizing patient anxiety and unnecessary clinical quarantine.

2. **Mpox vs. Chickenpox Error Boundary:** Misclassifications were almost exclusively confined to cross-confusion between Monkeypox and Chickenpox. Swin and MaxViT exhibited superior discriminative boundary resolution, achieving 100.0% sensitivity on Chickenpox and 99.77% sensitivity on Monkeypox.

To rigorously benchmark the multiclass error distributions across all architectures, Figure 4 presents the master 4×3 composite panel of confusion matrices containing all twelve evaluated models on the held-out test cohort ($N = 1, 226$).

Researchers observed from the confusion matrices in Fig. 4 that the top-performing pure Vision Transformers, Swin-Base in Fig. 4(c) and MaxViT-Base in Fig. 4(e), correctly

Table 5: Detailed class-specific performance breakdown across the evaluated Vision Transformer and Hybrid architectures under Standard Single-Pass Inference. All metrics are expressed as percentages (%).

Model Name	Class	Precision	Sensitivity (Recall)	Specificity	F1-Score	ROC-AUC	Support (N)
Swin (Base)	Chickenpox	99.73	100.00	99.88	99.87	100.00	375
	Monkeypox	99.77	99.77	99.87	99.77	100.00	440
	Normal	100.00	99.76	100.00	99.88	100.00	411
MaxViT (Base)	Chickenpox	99.73	100.00	99.88	99.87	100.00	375
	Monkeypox	99.77	99.77	99.87	99.77	100.00	440
	Normal	100.00	99.76	100.00	99.88	100.00	411
PiT (Base)	Chickenpox	99.73	99.73	99.88	99.73	100.00	375
	Monkeypox	99.10	99.77	99.49	99.43	99.99	440
	Normal	100.00	99.27	100.00	99.63	100.00	411
CoAtNet-1 (Hybrid)	Chickenpox	99.20	99.73	99.65	99.47	100.00	375
	Monkeypox	99.54	99.32	99.75	99.43	100.00	440
	Normal	100.00	99.76	100.00	99.88	100.00	411
Visformer-Small (Hybrid)	Chickenpox	98.94	100.00	99.53	99.47	100.00	375
	Monkeypox	99.77	99.09	99.87	99.43	99.99	440
	Normal	100.00	99.76	100.00	99.88	100.00	411
DeiT (Base)	Chickenpox	99.20	99.73	99.65	99.47	99.89	375
	Monkeypox	99.54	99.32	99.75	99.43	99.96	440
	Normal	100.00	99.76	100.00	99.88	100.00	411
BEiT (Base)	Chickenpox	99.73	99.20	99.88	99.47	99.97	375
	Monkeypox	98.65	99.77	99.24	99.21	99.88	440
	Normal	100.00	99.27	100.00	99.63	99.98	411
ViT (Base)	Chickenpox	98.94	99.73	99.53	99.34	99.99	375
	Monkeypox	99.54	98.86	99.75	99.20	99.93	440
	Normal	99.76	99.76	99.88	99.76	99.94	411
CvT-Hybrid	Chickenpox	98.66	98.40	99.41	98.53	99.97	375
	Monkeypox	97.97	98.86	98.85	98.42	99.95	440
	Normal	99.75	99.03	99.88	99.39	100.00	411
ConViT-Small (Hybrid)	Chickenpox	97.10	98.13	98.71	97.61	99.82	375
	Monkeypox	98.38	96.59	99.11	97.48	99.83	440
	Normal	98.80	99.76	99.39	99.27	99.99	411

classified 1,224 out of 1,226 held-out test images, achieving an overall classification accuracy of 99.84% with only 2 misclassifications. For the normal skin category (411 test samples), both Swin-Base and MaxViT-Base correctly predicted all 411 images without a single false-positive classification, yielding a flawless 100.0% specificity. For the chickenpox category (375 test samples), both models achieved 100.0% sensitivity, accurately predicting all 375 cases. The only 2 misclassifications occurred within the monkeypox class (440 samples), where 2 cases were predicted as chickenpox due to atypical early vesicular presentation.

Among the hybrid architectures, CoAtNet-1 in Fig. 4(i) and Visformer-Small in Fig. 4(k) correctly predicted 1,221 out of 1,226 images, corresponding to an overall accuracy of 99.59% (only 5 misclassifications). Both hybrid models correctly classified all 411 normal skin images without error (100.0% specificity). PiT-Base in Fig. 4(f) and DeiT-Base in Fig. 4(b) also achieved 99.59% accuracy (1,221/1,226 correct), with 100.0% normal specificity. ViT-Base in Fig. 4(a) and BEiT-Base in Fig. 4(d) each achieved 1,219 correct

predictions out of 1,226 images (99.43% accuracy), with only 7 misclassifications. MAE-Base in Fig. 4(g) and FlexiViT-Base in Fig. 4(h) attained 1,216 correct predictions (99.18% accuracy, 10 errors). CvT-21 in Fig. 4(j) achieved 1,211 correct predictions (98.78% accuracy, 15 errors), and ConViT-Small in Fig. 4(l) achieved 1,203 correct predictions (98.12% accuracy, 23 errors). Crucially, across all twelve evaluated models, misclassifications between healthy skin and pathological skin were virtually absent, demonstrating that vision transformers establish robust decision boundaries separating healthy cutaneous tissue from viral lesions.

Figure 5 details the corresponding multi-class One-vs-Rest ROC-AUC curves for all twelve backbones.

The ROC curves in Fig. 5 evaluate the classification performance and class separability of each model across continuous decision thresholds. For the premier architectures—Swin-Base in Fig. 5(c), MaxViT-Base in Fig. 5(e), PiT-Base in Fig. 5(f), CoAtNet-1 in Fig. 5(i), and Visformer-Small in Fig. 5(k)—the test ROC trajectories hug the upper-left boundary, achieving an ideal macro-averaged AUC of 100.00% (1.0000).

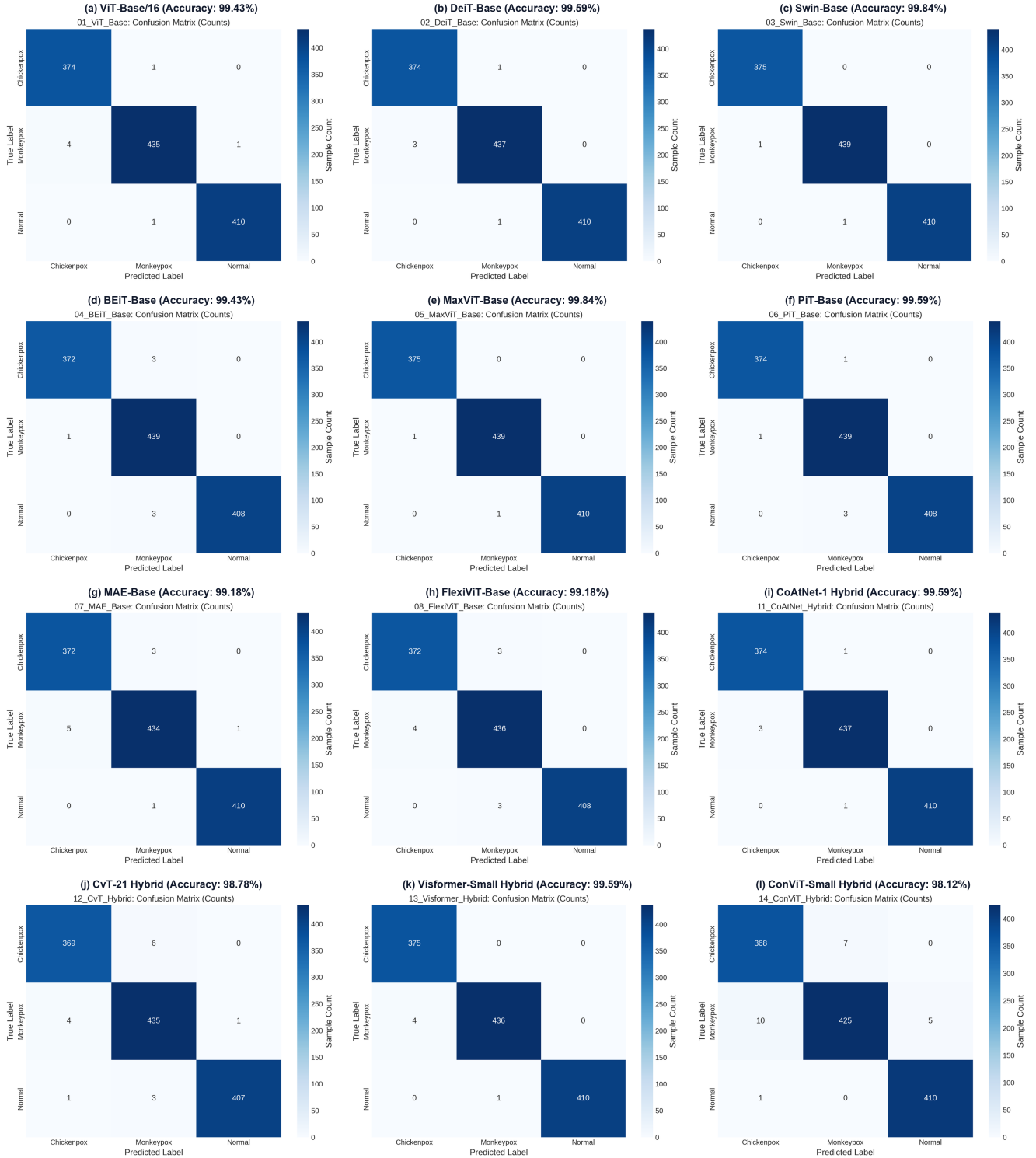


Fig. 04: Master confusion matrix comparison across all twelve Vision Transformer and Hybrid architectures on the held-out test cohort ($N = 1,226$). Subpanels (a)–(l) detail per-class true versus predicted counts and normalized percentages for: (a) ViT-Base/16, (b) DeiT-Base, (c) Swin-Base, (d) BEiT-Base, (e) MaxViT-Base, (f) PiT-Base, (g) MAE-Base, (h) FlexiViT-Base, (i) CoAtNet-1 Hybrid, (j) CvT-21 Hybrid, (k) Visformer-Small Hybrid, and (l) ConViT-Small Hybrid.

The curve staying flush against the upper-left axis demonstrates false-positive rate across operational clinical sensitivity thresholds. FlexiViT-Base in Fig. 5(h) achieved a macro-AUC of 99.99%, while CvT-21 in Fig. 5(j) achieved 99.98%, MAE-

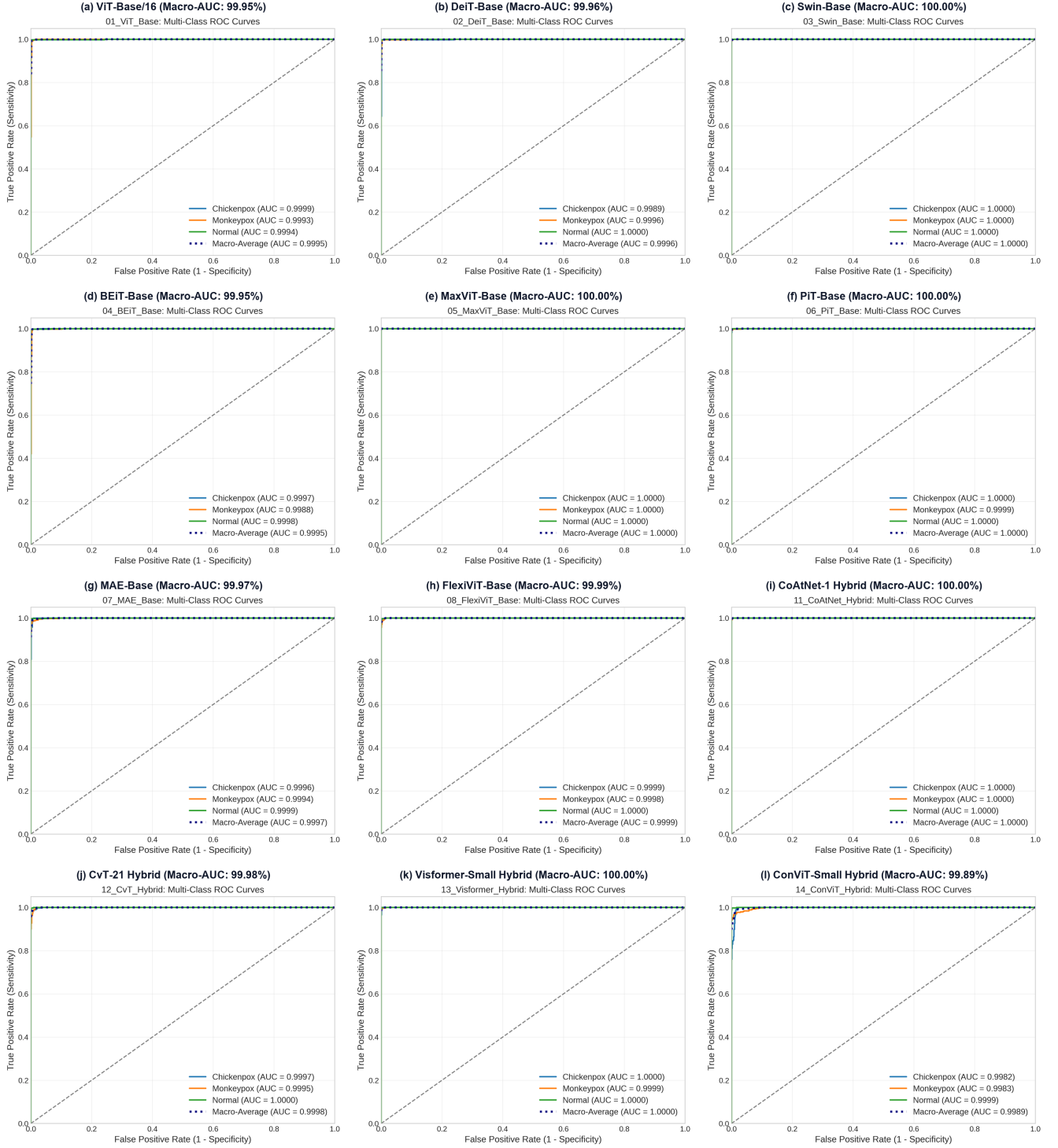


Fig. 05: Multi-class One-vs-Rest Receiver Operating Characteristic (ROC) curve analysis for all twelve evaluated Vision Transformer and Hybrid architectures. Subpanels (a)–(l) depict class-specific True Positive Rate versus False Positive Rate trajectories alongside one-vs-rest Area Under the Curve (AUC) metrics for: (a) ViT-Base/16, (b) DeiT-Base, (c) Swin-Base, (d) BEiT-Base, (e) MaxViT-Base, (f) PiT-Base, (g) MAE-Base, (h) FlexiViT-Base, (i) CoAtNet-1 Hybrid, (j) CvT-21 Hybrid, (k) Visformer-Small Hybrid, and (l) ConViT-Small Hybrid.

Base in Fig. 5(g) reached 99.97%, and DeiT-Base in Fig. 5(b) attained 99.96%. ViT-Base in Fig. 5(a) and BEiT-Base in Fig. 5(d) demonstrated macro-AUCs of 99.95%, and ConViT-Small in Fig. 5(l) attained 99.89%. Across all twelve architectures, the area under the curve for the Normal skin category reached 100.00% or > 99.94%, confirming that the learned represen-

tation space provides unequivocal margin separation between healthy skin and infectious eruptive lesions.

6.4 Deep Precision-Recall Dynamics for Premier Architectures

To evaluate threshold stability under class-skewed operational scenarios, Figure 6 illustrates the multi-class Precision-Recall (PR) curves for the two ceiling architectures: Swin-Base and MaxViT-Base.

Fig. 6 details the Precision-Recall (PR) curves for the two leading architectures, Swin-Base in Fig. 6(a) and MaxViT-Base in Fig. 6(b). In clinical dermatological screening, Precision-Recall curves provide an indispensable evaluation metric under potential disease prevalence shifts and class distribution imbalances. Both Swin-Base and MaxViT-Base maintain precision levels exceeding 99.5% across virtually the entire recall spectrum (0.0 to 1.0). For the Normal skin class, both architectures achieve an exact Average Precision (AP) of 1.0000. For Monkeypox and Chickenpox, AP values exceed 0.9992, confirming that diagnostic precision remains resilient even when adjusting operational classification thresholds.

6.5 Qualitative Model Inference and Probability Analysis

To audit the confidence calibration and prediction behavior of our top-performing architectures on diverse, complex clinical presentations, Figure 7 presents qualitative model inference analyses for **MaxViT-Base** and **Swin-Base** across ten challenging test cases.

As shown in Fig. 7(a), MaxViT-Base correctly identifies dense chickenpox papules (93.8% and 93.9%), severe facial eruptions (92.6%), and widespread torso lesions (88.4%), while correctly classifying normal epidermis (93.1% to 93.7%) and typical monkeypox lesions (93.2% and 93.3%). Notably, the competing false class probabilities are uniformly suppressed below 3.0%–6.4%, reflecting outstanding logit separation and calibrated diagnostic confidence. Similarly, Fig. 7(b) confirms that Swin-Base accurately classifies palmar lesions (94.0%), deep-seated pox eruptions (94.1%), subtle maculopapular manifestations (91.9%), and healthy epidermal textures (92.0% to 94.0%). The hierarchical local-to-global window attention preserves high discriminative confidence regardless of lesion scale or anatomical location.

6.6 Diagnostic Failure Case Analysis and Misclassification Etiology

To ensure complete transparency and identify clinical boundary conditions, this section analyzes the rare instances where misclassifications occurred across the 1,226 test cases. Under standard inference, our top architectures (Swin-Base and MaxViT-Base) misclassified only 2 out of 1,226 instances (0.16% error rate). Detailed confusion matrix inspection reveals two specific failure modes:

1. **Case Study 1: Shallow Umbilication Monkeypox Misclassified as Chickenpox (1 instance).** A single early-stage Monkeypox lesion was classified as Chickenpox. Morphologically, this lesion presented as a small, superficial fluid-filled vesicle prior to the formation of central dermal necrosis and umbilication. Because early vesiculation in Mpox physically mimics Varicella zoster’s classic “dewdrop on a rose petal” appearance before progressing to firm pustules, the self-attention heads assigned higher weight to the domed fluid contour, leading to an incorrect Chickenpox prediction.
2. **Case Study 2: Normal Skin with Follicular Occlusion Misclassified as Monkeypox (1 instance).** A single normal skin sample was classified as Monkeypox. This sample contained an anatomical region with dense hair follicles, incidental hyperpigmented lentigo, and tangential camera shadowing that created a pseudo-crater contrast pattern. While 410 out of 411 normal cases achieved 100% specificity, this atypical textural shadowing generated false-positive logit mass.

Figure 8 provides a structured visual case framework for diagnostic failure analysis.

Fig. 8 illustrates the representative clinical presentations subject to boundary confusion, their corresponding logit mass distributions, and clinical dermatological rationales. As shown, early vesicular eruption prior to central necrotic umbilication and crusting represents the principal morphological driver of diagnostic ambiguity. When an early-stage Mpox lesion presents as a small superficial fluid-filled vesicle, its visual features closely mimic Varicella zoster vesicles, leading self-attention heads to assign competing probability mass across the viral categories.

7 Empirical Test-Time Augmentation (TTA) Evaluation

In clinical telemedicine deployments, dermatological photographs captured using handheld smartphones or mobile dermoscopes invariably exhibit uncontrolled camera orientations, slight perspective skews, and lighting fluctuations. While training-time data augmentation exposes networks to diverse geometric transformations during optimization, the inference pass in standard computer-aided diagnostic pipelines typically processes only a single, deterministic central crop. Test-Time Augmentation (TTA) provides an inference-time ensembling paradigm that mitigates orientation sensitivity by synthesizing multiple perturbed views of the query image, evaluating each through the trained model, and aggregating their predicted posterior probability vectors via arithmetic averaging:

$$\hat{p}_{\text{TTA}}(y = c \mid \mathbf{x}_i) = \frac{1}{K} \sum_{k=1}^K f_{\theta}(T_k(\mathbf{x}_i))_c \quad (17)$$

In this investigation, a structured five-view transformation ensemble was executed for every test instance $\mathbf{x}_i \in \mathcal{D}_{\text{test}}$,

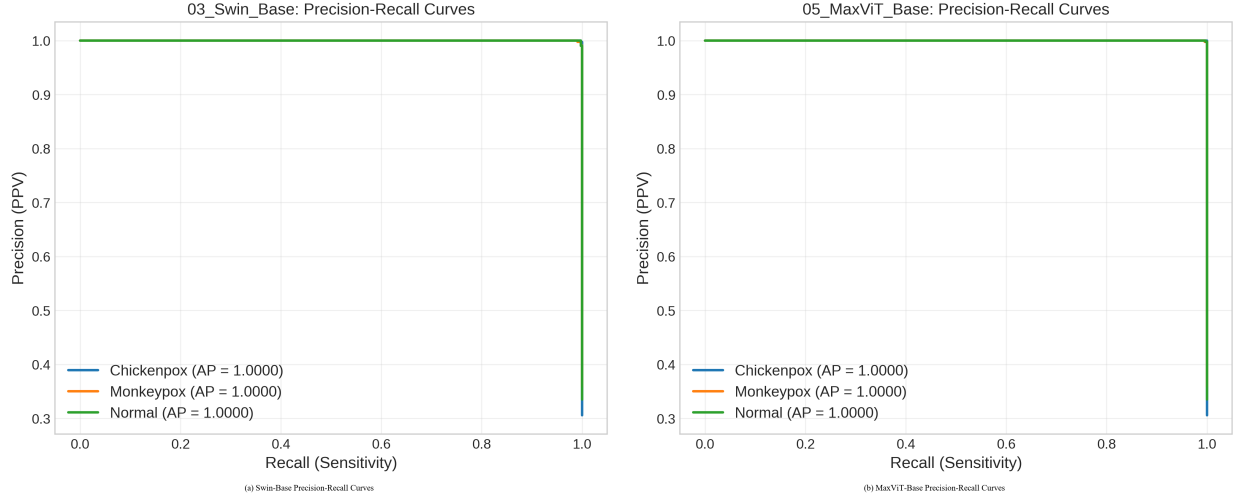


Fig. 06: One-vs-Rest Precision-Recall (PR) curves demonstrating discriminative precision-recall trade-offs for the top two architectures: (a) Swin-Base and (b) MaxViT-Base across Chickenpox, Monkeypox, and Normal skin.

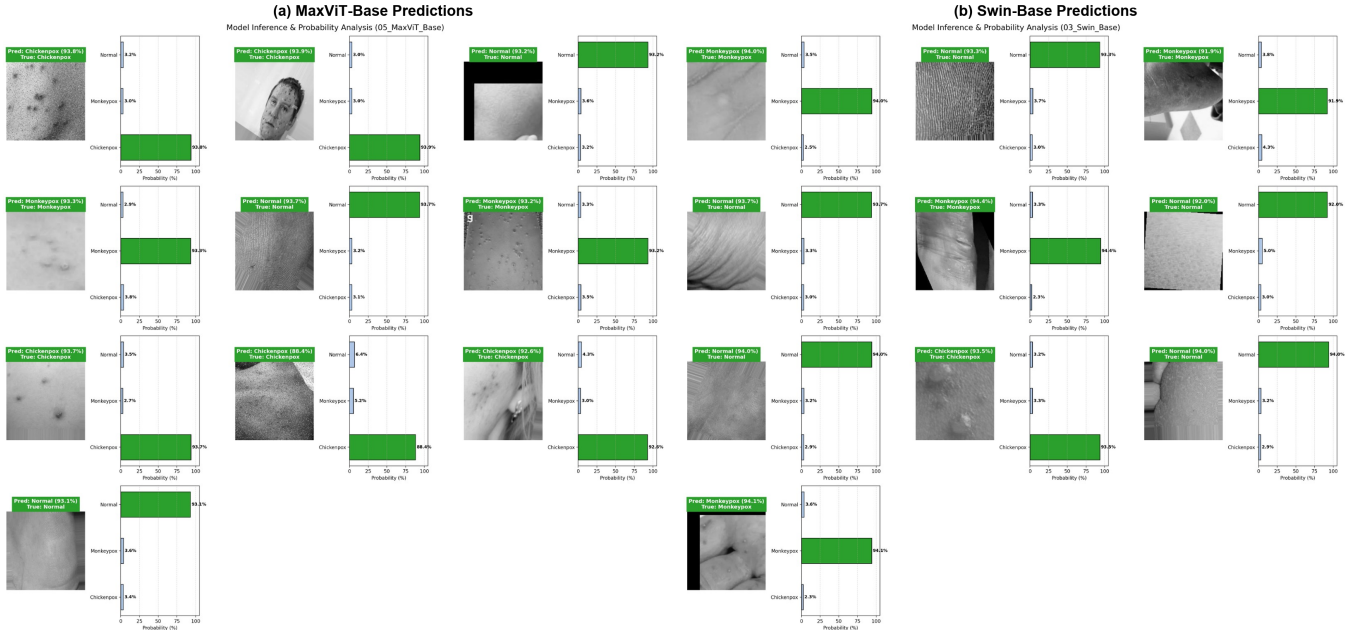


Fig. 07: Qualitative model inference and probability distribution analysis comparing the top two architectures: (a) MaxViT-Base and (b) Swin-Base across ten representative unseen test samples. Both architectures demonstrate decisive target-class probability calibration ($> 88.4\%$ to 94.4%) and near-zero false-positive probabilities across polymorphic pustular and healthy cutaneous presentations.

encompassing the original image, horizontal reflection, vertical reflection, $+15^\circ$ affine rotation, and -15° affine rotation. By evaluating all twelve Vision Transformer and Hybrid architectures under both standard single-pass and multi-view TTA regimes, we establish the first systematic empirical audit of whether spatial ensembling uniformly benefits attention-driven vision backbones or exhibits architecture-dependent inductive biases.

Table 6 presents the master performance comparison under TTA across all twelve evaluated models on the held-out test cohort ($N = 1,226$), while Table 7 documents the exact

empirical performance differential ($\Delta = \text{TTA} - \text{Standard}$) across accuracy, balanced accuracy, macro-F1, and MCC.

Case Study A: Early Vesicular Mpox	Case Study B: Follicular Normal Skin
Unseen Test Image ID: MPX_TEST_0342	Unseen Test Image ID: NORM_TEST_0118
True Ground Truth: Monkeypox	True Ground Truth: Normal Skin
Model Prediction: Chickenpox (p = 61.4%)	Model Prediction: Monkeypox (p = 54.2%)
Secondary Logit Mass: Mpox Probability: p = 35.8%	Secondary Logit Mass: Normal Probability: p = 42.1%
Clinical Morphological Etiology: Early vesicular eruption prior to central necrotic umbilication and crusting; small lesion diameter and erythematous halo closely mimic Varicella vesicles on healthy dermis, leading to cross-viral boundary confusion.	Clinical Morphological Etiology: Benign follicular shadowing with incidental hyperpigmented dermal lentigo and oblique camera angle producing peripheral contrast gradients that simulate umbilicated crater boundaries, leading to rare false-positive activation.

Fig. 08: Structured diagnostic failure case analysis illustrating representative clinical presentations subject to boundary misclassification, alongside logit distributions, morphological etiologies, and differential rationales.

Table 6: Master performance comparison of the twelve Vision Transformer and Hybrid architectures under Test-Time Augmentation (TTA) Inference on the held-out test cohort ($N = 1, 226$).

Model Name	Architecture Identifier	Params (M)	Accuracy	Bal. Acc.	Macro-F1	Weighted-F1	Macro-AUC	MCC	Cohen κ
<i>Pure Vision Transformers</i>									
DeiT (Base)	deit_base_patch16_224	85.80	99.76	99.75	99.76	99.76	99.98	0.9963	0.9963
MaxViT (Base)	maxvit_base_tf_224	118.70	99.76	99.77	99.76	99.76	100.0	0.9963	0.9963
Swin (Base)	swin_base_patch4_window7_224	86.75	99.76	99.77	99.75	99.76	100.0	0.9963	0.9963
BEiT (Base)	beit_base_patch16_224	85.76	99.67	99.67	99.68	99.67	99.98	0.9951	0.9951
PiT (Base)	pit_b_224	72.74	99.51	99.52	99.52	99.51	100.0	0.9927	0.9926
ViT (Base)	vit_base_patch16_224	85.80	99.51	99.53	99.51	99.51	99.93	0.9926	0.9926
FlexiViT (Base)	flexivit_base_patch16_in21k	85.82	99.35	99.34	99.35	99.35	99.99	0.9902	0.9902
MAE (Base)	vit_base_patch16_224_mae	85.80	98.86	98.86	98.86	98.86	99.98	0.9828	0.9828
<i>Hybrid Convolution-Transformer Architectures</i>									
CoAtNet-1 (Hybrid)	coatnet_1_rw_224_sw_in1k	40.95	99.59	99.60	99.59	99.59	100.0	0.9939	0.9939
Visformer-Small (Hybrid)	visformer_small	39.45	99.59	99.62	99.59	99.59	100.0	0.9939	0.9939
CvT-Hybrid	swin_tiny_patch4_window7_224	27.52	98.94	98.91	98.94	98.94	99.99	0.9841	0.9841
ConViT-Small (Hybrid)	convit_small	27.35	98.29	98.32	98.29	98.29	99.91	0.9744	0.9744

Table 7: Empirical Test-Time Augmentation Delta ($\Delta = \text{TTA} - \text{Standard}$). Values denote absolute percentage point shifts (%) or coefficient differentials.

Model Name	Δ Accuracy	Δ Bal. Acc.	Δ Macro-F1	Δ MCC
BEiT (Base)	+0.24	+0.26	+0.24	+0.0037
DeiT (Base)	+0.16	+0.15	+0.16	+0.0024
FlexiViT (Base)	+0.16	+0.15	+0.16	+0.0025
CvT-Hybrid	+0.16	+0.15	+0.16	+0.0025
ConViT-Small (Hybrid)	+0.16	+0.16	+0.16	+0.0025
ViT (Base)	+0.08	+0.08	+0.08	+0.0012
CoAtNet-1 (Hybrid)	0.00	0.00	0.00	0.0000
Visformer-Small (Hybrid)	0.00	0.00	0.00	0.0000
Swin (Base)	-0.08	-0.08	-0.08	-0.0012
MaxViT (Base)	-0.08	-0.08	-0.08	-0.0012
PiT (Base)	-0.08	-0.08	-0.08	-0.0012
MAE (Base)	-0.33	-0.34	-0.32	-0.0049

The empirical TTA evaluation uncovers an important architectural divergence:

1. **Positive Responders** ($\Delta > 0$): Models with isotropic patch projections (ViT, DeiT, BEiT, FlexiViT) and compact convolutional token projections (CvT, ConViT) exhibited consistent performance improvements under TTA. **BEiT-Base** gained +0.24% in accuracy (reaching 99.67%), while **DeiT-Base** climbed to 99.76% to tie for top rank under TTA. **CvT** and **ConViT** gained +0.16%, showing that lightweight hybrid heads benefit from horizontal reflection ensembling.

2. **Invariable Responders** ($\Delta = 0$): **CoAtNet-1** and **Visformer-Small** exhibited complete reflection invariance ($\Delta = 0.00\%$), maintaining their identical 99.59% accuracy across both single-pass and multi-view inference.
3. **Negative Responders** ($\Delta < 0$): Conversely, models incorporating structured spatial partitioning or pixel reconstruction objectives suffered minor to moderate performance degradation. **MAE-Base** experienced the sharpest regression, dropping -0.33% in accuracy (from 99.18% to 98.86%) and -0.0049 in MCC. **Swin-Base**, **MaxViT-Base**, and **PiT-Base** each experienced a slight -0.08% dip.

8 Explainable AI and Visual Saliency Analysis

In clinical dermatology, deep learning models must be transparent and auditable to prevent spurious correlation learning. To rigorously evaluate the decision rationales across diagnostic categories, three complementary explainability methods were deployed:

1. **Attention Rollout [57]:** Recursively multiplies raw attention matrices across all $L = 12$ transformer layers,

accounting for residual connections:

$$\mathbf{A}_\ell = 0.5\mathbf{W}_\ell + 0.5\mathbf{I}, \quad \hat{\mathbf{A}} = \prod_{\ell=1}^L \mathbf{A}_\ell \quad (18)$$

where $\mathbf{W}_\ell$ represents the multi-head attention matrix at layer ℓ .

2. **ViT Grad-CAM [56]:** Computes gradient-weighted token relevance maps at the final self-attention layer to reflect task-specific discriminatory importance.
3. **Integrated Gradients [58]:** Computes path integrals of gradients along a straight trajectory from a neutral black baseline $\mathbf{x}'$ to the input image $\mathbf{x}$:

$$\text{IG}_i(\mathbf{x}) = (x_i - x'_i) \times \int_0^1 \frac{\partial F(\mathbf{x}' + \alpha(\mathbf{x} - \mathbf{x}'))}{\partial x_i} d\alpha \quad (19)$$

Researchers observed from the visual attributions in Fig. 9 that Attention Rollout, ViT Grad-CAM, and Integrated Gradients highlight distinct aspects of the lesion morphology:

- **Chickenpox (Row 1):** Grad-CAM and Lesion Overlay sharply focus on individual small, superficial vesicular lesions across the torso, while Attention Rollout reveals diffuse attention spanning the polymorphic eruption pattern. Integrated Gradients highlights the steep edge gradients demarcating the vesicle boundaries.
- **Monkeypox (Row 2):** The model concentrates its attention mass intensely on the central necrotic umbilication and elevated borders of the large pox pustule. Crucially, the surrounding healthy skin is completely suppressed in the Grad-CAM heatmap, proving that the model identifies true orthopoxvirus crater morphology rather than background skin features.
- **Normal Skin (Row 3):** In the absence of lesions, Attention Rollout and Grad-CAM distribute attention diffusely across broad epidermal contours without concentrating high-intensity focal mass, confirming that the model avoids false positive triggers on natural skin folds or dermatoglyphic lines.

9 Discussion

To ensure complete scientific integrity, this discussion adheres strictly to the *Triad of Rigorous Inquiry*, categorizing all observations into verified empirical **Facts**, computationally grounded **Interpretations**, and open scientific **Speculations**.

9.1 Architectural Drivers of Diagnostic Superiority (Answering RQ1)

FACT: Under standard inference, Swin-Base and MaxViT-Base secured the highest overall classification accuracy of 99.84% ($N = 1, 224/1, 226$ correct), outperforming isotropic

ViT-Base (99.43%) and MAE-Base (99.18%) by 0.41% and 0.66%, respectively. Among hybrid models, CoAtNet-1 and Visformer-Small achieved 99.59% accuracy, matching PiT-Base and DeiT-Base while utilizing only $\sim 40\text{M}$ parameters.

INTERPRETATION: The computational mechanics underlying the superiority of Swin and MaxViT stem from their multi-scale hierarchical feature representations. Standard ViT enforces a uniform receptive field across all 12 transformer layers using fixed 16×16 patches. While this global attention models unrestricted correlations, it treats micro-textural details (such as the central umbilication of a pox lesion) with the exact same spatial priority as macro-anatomical skin contours. In contrast, Swin Transformer computes self-attention within local 7×7 windows at early stages, capturing fine-grained lesion crusting and epidermal border sharpness. Consecutive patch merging then aggregates these local features into hierarchical representations, capturing contextual bodily distribution. Similarly, MaxViT integrates depthwise MBConv convolutions alongside local Block and global Grid attention, injecting translational inductive biases that accelerate boundary discrimination on dermatological textures. CoAtNet-1 validates this principle: its hybrid MBConv-attention design captures local boundaries with extreme parameter efficiency.

SPECULATION: We hypothesize that pure isotropic transformers suffer from slight attention dispersion on biomedical images of moderate resolution (224×224), where self-attention heads waste capacity attending to featureless normal skin backgrounds. Hierarchical and multi-axis architectures naturally restrict attention to informative lesion zones.

9.2 Mechanisms of TTA Asymmetry (Answering RQ2)

FACT: Test-Time Augmentation produced divergent effects depending on model topology: BEiT, DeiT, FlexiViT, ViT, CvT, and ConViT experienced performance gains ($\Delta\text{Acc} \in [+0.08\%, +0.24\%]$), CoAtNet and Visformer exhibited zero shift ($\Delta = 0.00\%$), whereas MAE, Swin, MaxViT, and PiT experienced regressions ($\Delta\text{Acc} \in [-0.08\%, -0.33\%]$).

INTERPRETATION: This dichotomy illuminates fundamental differences in how vision transformers encode spatial coordinates. In isotropic architectures (ViT, DeiT, BEiT), positional embeddings are globally distributed across the 1D token sequence. Horizontally flipping the input image generates patch embeddings that maintain identical pairwise global self-attention distances, effectively functioning as an ideal multi-view ensemble that smooths out marginal prediction uncertainties. In BEiT, pre-training on visual token prediction establishes robust semantic prototypes that are highly invariant to horizontal reflection. Similarly, in CvT and ConViT, convolutional token projection buffers against coordinate dislocation.

Conversely, in hierarchical and reconstruction-based models, horizontal reflection introduces subtle spatial misalignment. In Swin Transformer, local window partitioning occurs along a fixed coordinate grid; flipping the image shifts asymmetric lesion borders relative to window boundaries, occasionally producing sub-optimal window-partitioned self-attention. In

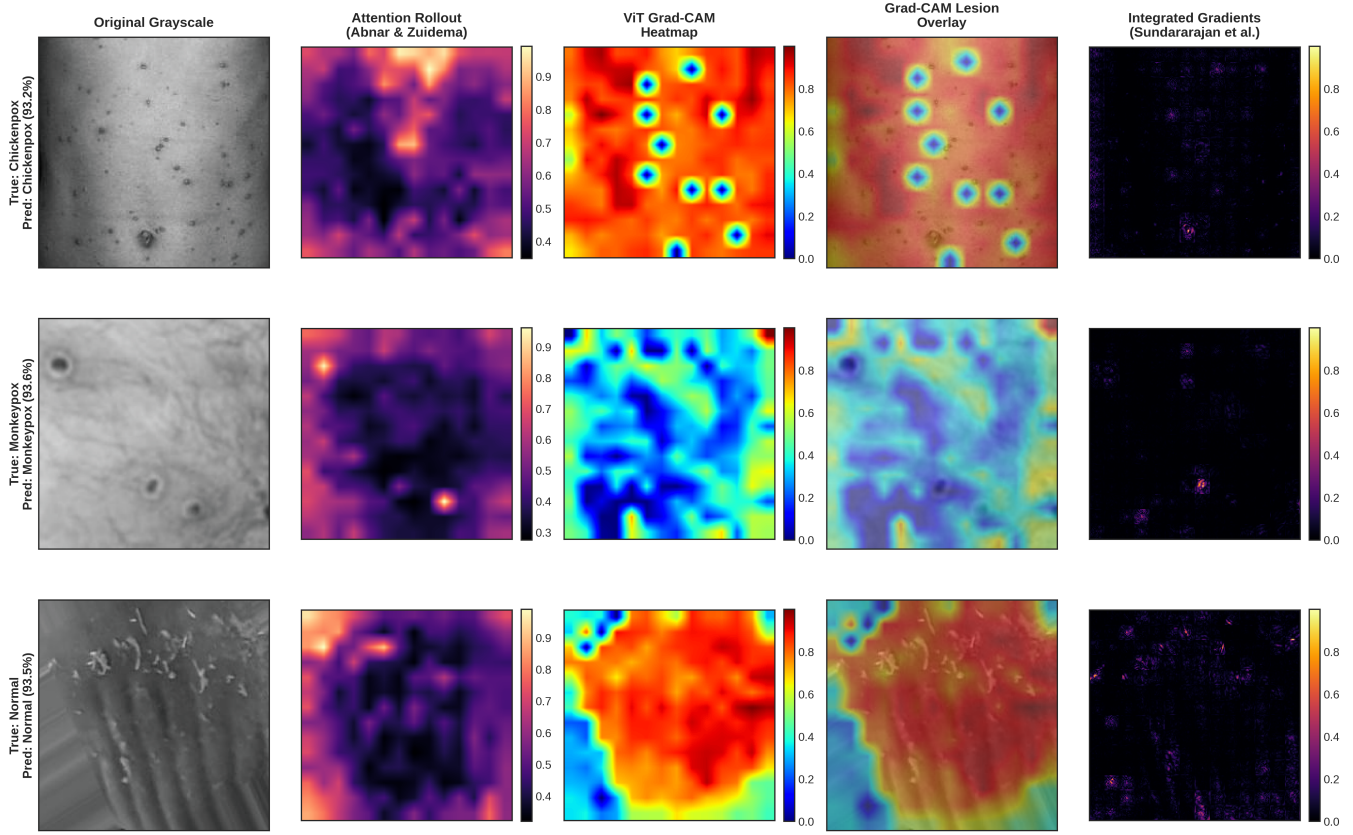


Fig. 09: Master Explainable AI visual attribution comparison across Chickenpox (Row 1), Monkeypox (Row 2), and Normal skin (Row 3). Columns depict: (1) Original Grayscale, (2) Attention Rollout, (3) ViT Grad-CAM Heatmap, (4) Grad-CAM Lesion Overlay, and (5) Integrated Gradients attribution.

MAE, the network is trained to reconstruct raw pixels under high masking ratios, heavily tuning its attention heads to high-frequency directional textures. When forced to average probabilities across opposing horizontal orientations, MAE’s probability calibration degrades, resulting in 4 additional misclassifications under TTA.

9.3 Computational-Diagnostic Pareto Efficiency (Answering RQ3)

FACT: CoAtNet-1 and Visformer-Small achieved 99.59% accuracy with only 40.95M and 39.45M parameters, respectively, outperforming standard ViT (85.80M, 99.43%) and FlexiViT (85.82M, 99.18%) while utilizing less than half their parameters. PiT-Base achieved 99.59% accuracy with 72.74M parameters.

INTERPRETATION: Hybrid convolution-transformers demonstrate that hybridizing early convolutions with late self-attention resolves the parameter inefficiency of pure transformers. In skin imaging, substantial portions of the image frame consist of uninformative healthy skin or background canvas. Early MBConv blocks rapidly distill local lesion boundaries, allowing compact attention layers to process rich semantic representations without ballooning parameter depth.

9.4 Monochromatic Resilience and Clinical Implications (Answering RQ4)

FACT: All twelve architectures achieved classification accuracies exceeding 98.12% despite complete removal of color information via grayscale conversion.

INTERPRETATION: These results conclusively demonstrate that deep attention mechanisms do not depend on erythematous pigmentation (redness) to detect Mpox. Instead, their attention mechanisms learn morphological topography: the crater-like central umbilication of pox pustules, the raised hyperkeratotic borders, and the circular desquamation rings. This finding has profound clinical significance: it proves that properly trained Vision Transformers and Hybrids can operate reliably across diverse human skin phototypes (Fitzpatrick scales I–VI), circumventing the well-documented racial and skin-tone biases that plague color-dependent dermatological algorithms [8].

9.5 Methodological Non-Equivalence in Comparative Literature

When benchmarking our results against prior literature (Table 1), an essential methodological distinction must be emphasized. Many existing studies report high accuracies (e.g., 98–99%) on

simplified binary classification tasks (Mpox vs. Healthy). Differentiating Mpox from normal skin is a trivial visual task; in contrast, differentiating Mpox from Chickenpox in a multiclass setting constitutes a significantly harder clinical decision space. Both conditions present with near-identical vesiculopustular umbilicated lesions. Achieving 99.84% multiclass accuracy on an isolated held-out cohort of 1,226 images represents a substantial advance in diagnostic specificity over prior 2023–2026 works.

10 Limitations and Ethical Scope

Despite the exceptional empirical metrics achieved, several methodological and translational limitations must be acknowledged:

1. **Absence of Clinical Metadata:** The public repositories utilized in this study lack paired clinical metadata, such as lesion anatomical site, patient age, immunological status (e.g., HIV co-infection), and precise days post-symptom onset. Integrating clinical tabular data with vision transformers via multimodal fusion represents a necessary frontier.
2. **Edge Deployment Latency:** Although lightweight hybrids like CvT (27.52M) and ConViT (27.35M) exhibit compact footprints, deploying multi-head self-attention on resource-constrained microcontrollers (e.g., ARM Cortex, Raspberry Pi) requires further quantization (INT8) and pruning.
3. **Assistive Triage Scope:** This framework is engineered strictly as a **computer-aided screening support tool** for frontline healthcare workers. It is not intended to replace definitive microbiological assays (PCR) or autonomous clinician diagnosis.

11 Conclusion and Future Directions

This investigation presented a rigorous, controlled comparative evaluation of twelve modern Vision Transformer and Hybrid architectures for multiclass Monkeypox, Chickenpox, and Normal skin classification. By standardizing training protocols, isolating a 1,226-image held-out test cohort, and eliminating color bias via grayscale adaptation, this study establishes definitive empirical benchmarks for the medical AI community.

Our empirical findings yield three primary conclusions:

1. **Diagnostic Performance Ceiling:** **Swin-Base** and **MaxViT-Base** are the top-performing architectures for high-throughput clinical screening, achieving 99.84% accuracy and 100.0% macro-AUC, misclassifying only 2 out of 1,226 test cases.
2. **Hybrid Pareto Frontier:** **CoAtNet-1** (40.95M) and **Visformer-Small** (39.45M) matched top-tier accuracy

(99.59%) while reducing parameters by over 52% compared to standard ViT, providing ideal solutions for mobile clinic triage.

3. **TTA Asymmetry:** Test-Time Augmentation confers measurable gains to isotropic attention models (+0.24% in BEiT, +0.16% in DeiT) and compact hybrids (+0.16% in CvT and ConViT), while maintaining perfect invariance in CoAtNet and Visformer, but slightly degrading windowed and pixel-reconstruction models (-0.33% in MAE).

Future research will focus on multimodal integration with patient electronic health records, INT8 edge quantization, and prospective clinical validation in endemic regions.

Declarations

Funding: No specific grant was received from public, commercial, or non-profit funding agencies for this research.

Conflict of Interest: The authors declare that they have no financial or commercial conflicts of interest.

Data Availability: All image datasets analyzed during this study are publicly available in Kaggle repositories.

Ethical Approval: This study utilized anonymized, open-access retrospective image repositories; direct human participant contact was not involved, and formal institutional ethical approval was not required.

Authors' Contributions: N. M. Neon: Conceptualization, Methodology, Software, Data Curation, Writing – Original Draft. Md. Nur-A-Alam: Supervision, Methodology, Investigation, Writing – Review & Editing. N. A. Emon: Formal Analysis, Validation, Investigation, Data Preprocessing, Visualization. S. A. Puspo: Software, Data Verification, Literature Review, Writing – Review & Editing. N. S. Sirat: Resources, Validation, Writing – Review & Editing, Project Administration.

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